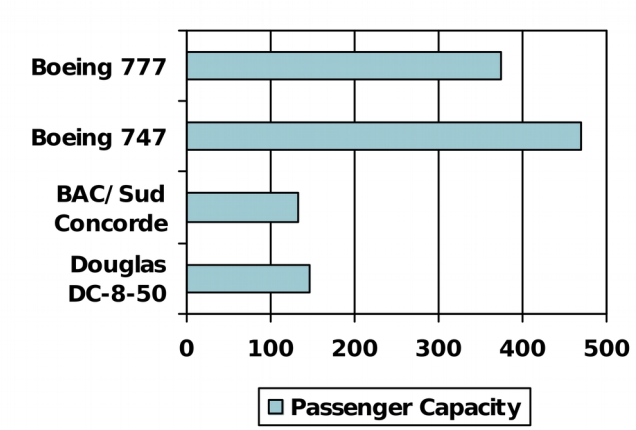
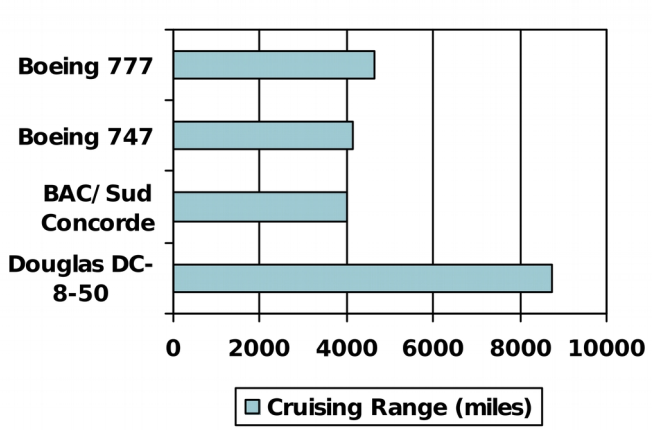
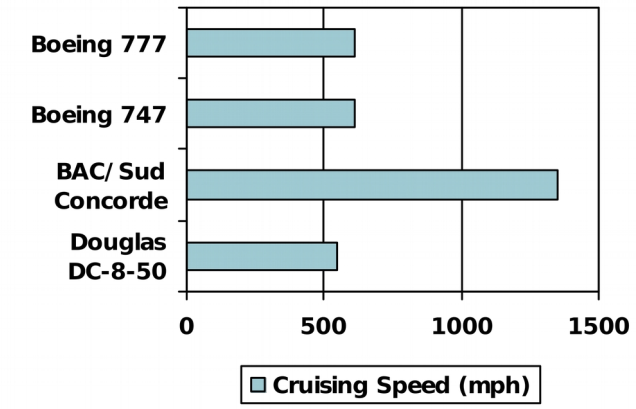
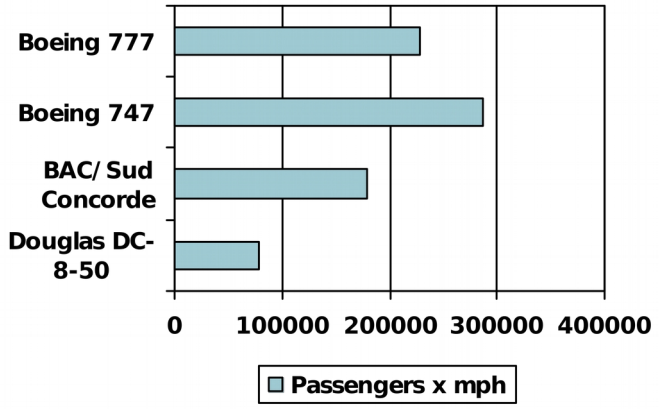
Defining Performance

- Which airplane has the best performance?









Response Time and Throughput

- Response time
  - How long it takes to do a task
- Throughput
  - Total work done per unit time
    - e.g., tasks/transactions/… per hour
- How are response time and throughput affected by
  - Replacing the processor with a faster version?
  - Adding more processors?
- We'll focus on response time for now…

Relative Performance

- Define Performance = 1/Execution Time
- "X is *n* time faster than Y"

$$\text{Performance}_X / \text{Performance}_Y$$

$$\text{Execution time}_Y / \text{Execution time}_X = n$$

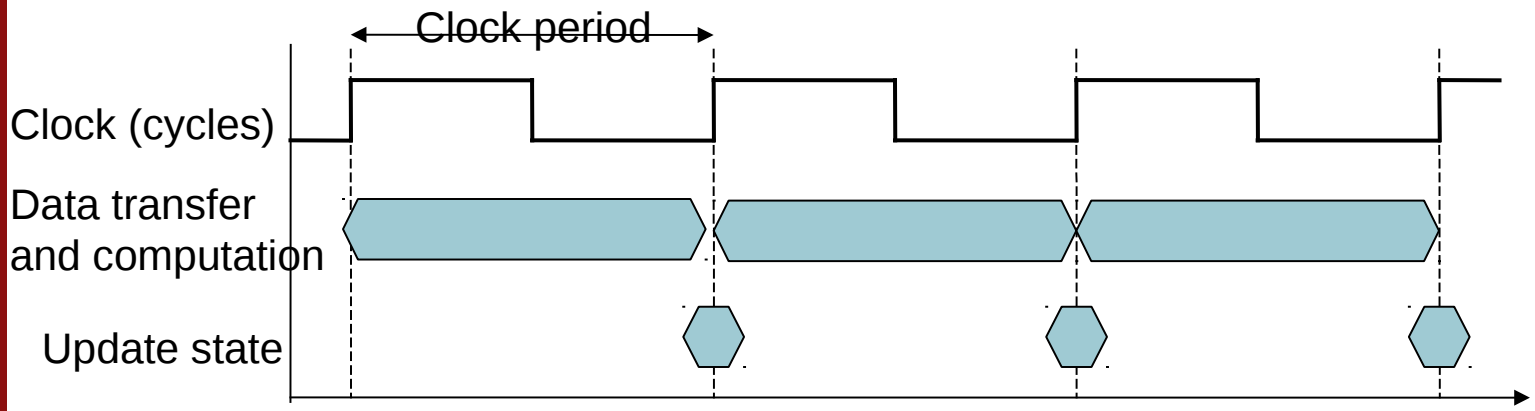
- Example: time taken to run a program
  - 10s on A, 15s on B
  - Execution Time_B / Execution Time_A
    = 15s / 10s = 1.5
  - So A is 1.5 times faster than B

Measuring Execution Time

- Elapsed time
  - Total response time, including all aspects
    - Processing, I/O, OS overhead, idle time
  - Determines system performance
- CPU time
  - Time spent processing a given job
    - Discounts I/O time, other jobs' shares
  - Comprises user CPU time and system CPU time
  - Different programs are affected differently by CPU and system performance

CPU Clocking

- Operation of digital hardware governed by a constant-rate clock



  - Clock period: duration of a clock cycle
    - e.g., 250ps = 0.25ns = 250×10^{-12}s
  - Clock frequency (rate): cycles per second
    - e.g., 4.0GHz = 4000MHz = 4.0×10^{9}Hz

CPU Time

$$\text{CPU Time} = \text{CPU Clock Cycles} \times \text{Clock Cycle Time}$$

$$\frac{\text{CPU Clock Cycles}}{\text{Clock Rate}}$$

- Performance improved by
  - Reducing number of clock cycles
  - Increasing clock rate
  - Hardware designer must often trade off clock rate against cycle count

CPU Time Example

- Computer A: 2GHz clock, 10s CPU time

Designing Computer B

- Aim for 6s CPU time
- Can do faster clock, but causes 1.2 × clock cycles

How fast must Computer B clock be?

CPU Time Example

- Computer A: 2GHz clock, 10s CPU time

  Designing Computer B
  - Aim for 6s CPU time
  - Can do faster clock, but causes 1.2 × clock cycles

  How fast must Computer B clock be?

$$\text{Clock Rate}_B = \frac{\text{Clock Cycles}_B}{\text{CPU Time}_B} = \frac{1.2 \times \text{Clock Cycles}_A}{6\text{s}}$$

$$\text{Clock Cycles}_A = \text{CPU Time}_A \times \text{Clock Rate}_A$$

$$10\text{s} \times 2\text{GHz} = 20 \times 10^9$$

$$\text{Clock Rate}_B = \frac{1.2 \times 20 \times 10^9}{6\text{s}} = \frac{24 \times 10^9}{6\text{s}} = 4\text{GHz}$$

Instruction Count and CPI

$$\text{Clock Cycles} = \text{Instruction Count} \times \text{Cycles per Instruction}$$

$$\text{CPU Time} = \text{Instruction Count} \times \text{CPI} \times \text{Clock Cycle Time}$$

$$\frac{\text{Instruction Count} \times \text{CPI}}{\text{Clock Rate}}$$

- Instruction Count for a program
  - Determined by program, ISA and compiler
- Average cycles per instruction
  - Determined by CPU hardware
  - If different instructions have different CPI
    - Average CPI affected by instruction mix

CPI Example

- Computer A: Cycle Time = 250ps, CPI = 2.0
- Computer B: Cycle Time = 500ps, CPI = 1.2
- Same ISA
- Which is faster, and by how much?

CPI Example

- Computer A: Cycle Time = 250ps, CPI = 2.0
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$$\text{CPU Time}_A = \text{Instruction Count} \times \text{CPI}_A \times \text{Cycle Time}_A$$

$$= I \times 2.0 \times 250\text{ps} = I \times 500\text{ps}$$

A is faster…

$$\text{CPU Time}_B = \text{Instruction Count} \times \text{CPI}_B \times \text{Cycle Time}_B$$

$$= I \times 1.2 \times 500\text{ps} = I \times 600\text{ps}$$

$$\frac{\text{CPU Time}_B}{\text{CPU Time}_A} = \frac{I \times 600\text{ps}}{I \times 500\text{ps}} = 1.2$$

…by this much

CPI in More Detail

- If different instruction classes take different numbers of cycles

$$\text{Clock Cycles} = \sum_{i=1}^{n} \left(\text{CPI}_i \times \text{Instruction Count}_i \right)$$

- Weighted average CPI

$$\text{CPI} = \frac{\text{Clock Cycles}}{\text{Instruction Count}} = \sum_{i=1}^{n} \left(\text{CPI}_i \times \underbrace{\frac{\text{Instruction Count}_i}{\text{Instruction Count}}}_{\text{Relative frequency}} \right)$$

CPI Example

- Alternative compiled code sequences using instructions in classes A, B, C

Class	A	B	C
CPI for class	1	2	3
IC in sequence 1	2	1	2
IC in sequence 2	4	1	1

- Average CPI for each sequence?

CPI Example

- Alternative compiled code sequences using instructions in classes A, B, C

Class	A	B	C
CPI for class	1	2	3
IC in sequence 1	2	1	2
IC in sequence 2	4	1	1

Sequence 1: IC = 5

- Clock Cycles
  = 2×1 + 1×2 + 2×3
  = 10
- Avg. CPI = 10/5 = 2.0

- Sequence 2: IC = 6
  - Clock Cycles
    = 4×1 + 1×2 + 1×3
    = 9
  - Avg. CPI = 9/6 = 1.5

Performance Summary

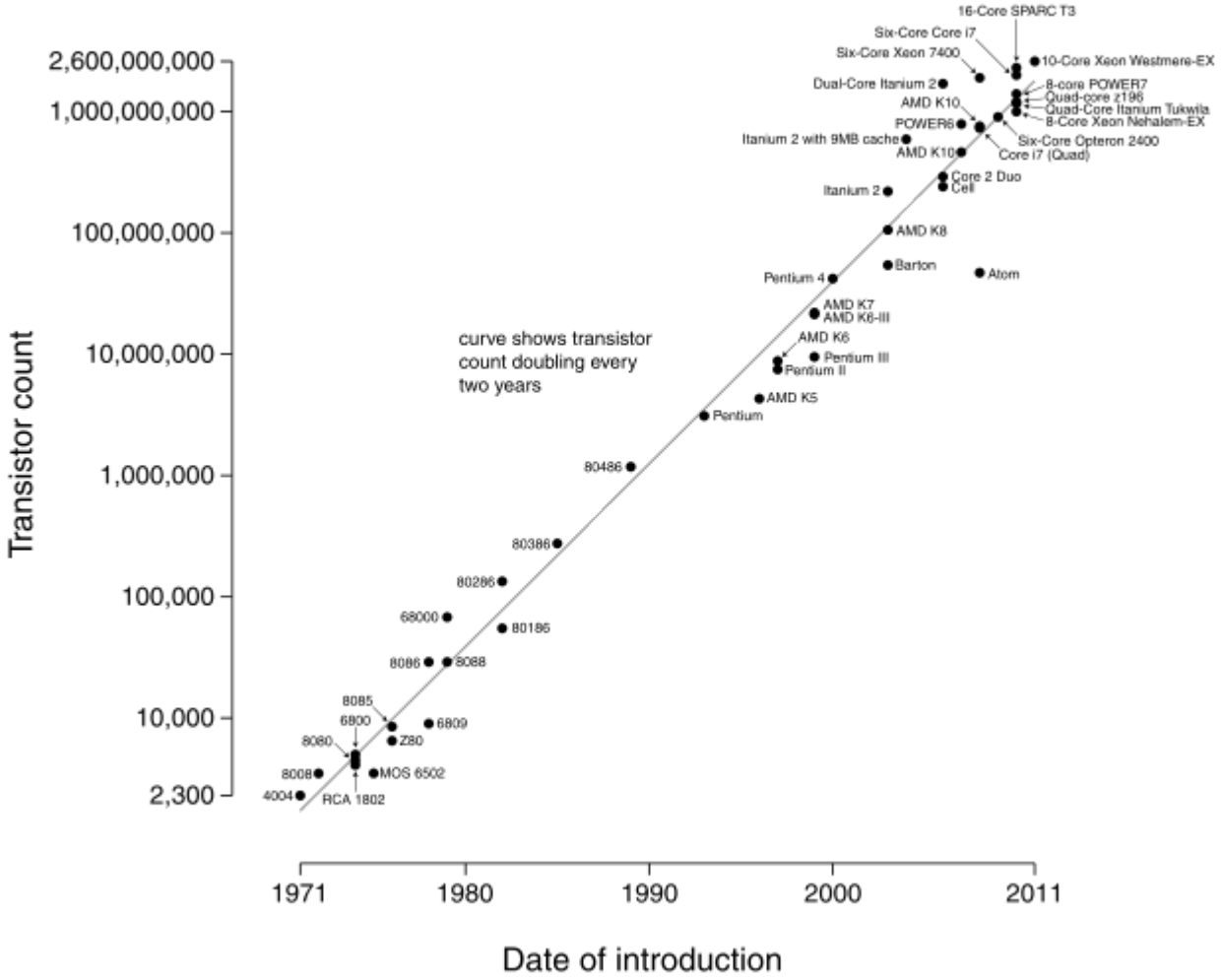
$$\text{CPU Time} = \frac{\text{Instructions}}{\text{Program}} \times \frac{\text{Clock cycles}}{\text{Instruction}} \times \frac{\text{Seconds}}{\text{Clock cycle}}$$

- Performance depends on
  - Algorithm: affects IC, possibly CPI
  - Programming language: affects IC, CPI
  - Compiler: affects IC, CPI
  - Instruction set architecture: affects IC, CPI, T_c

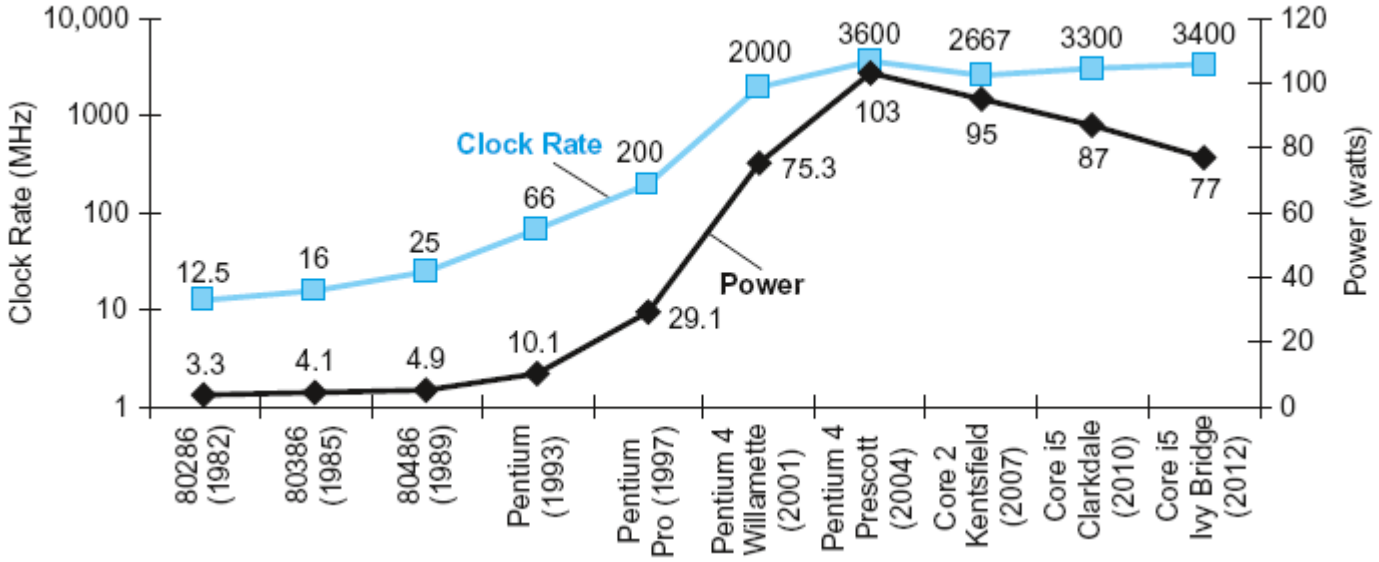
Moore’s Law

- Progress in computer technology Underpinned by Moore’s Law
  - Describes a long-term trend in computer hardware
  - The number of transistors that can be placed inexpensively on an integrated circuit has doubled approximately every two years
  - Is Moore's law really a law?

Microprocessor Transistor Counts 1971-2011 & Moore's Law



Power Trends



- In CMOS IC technology

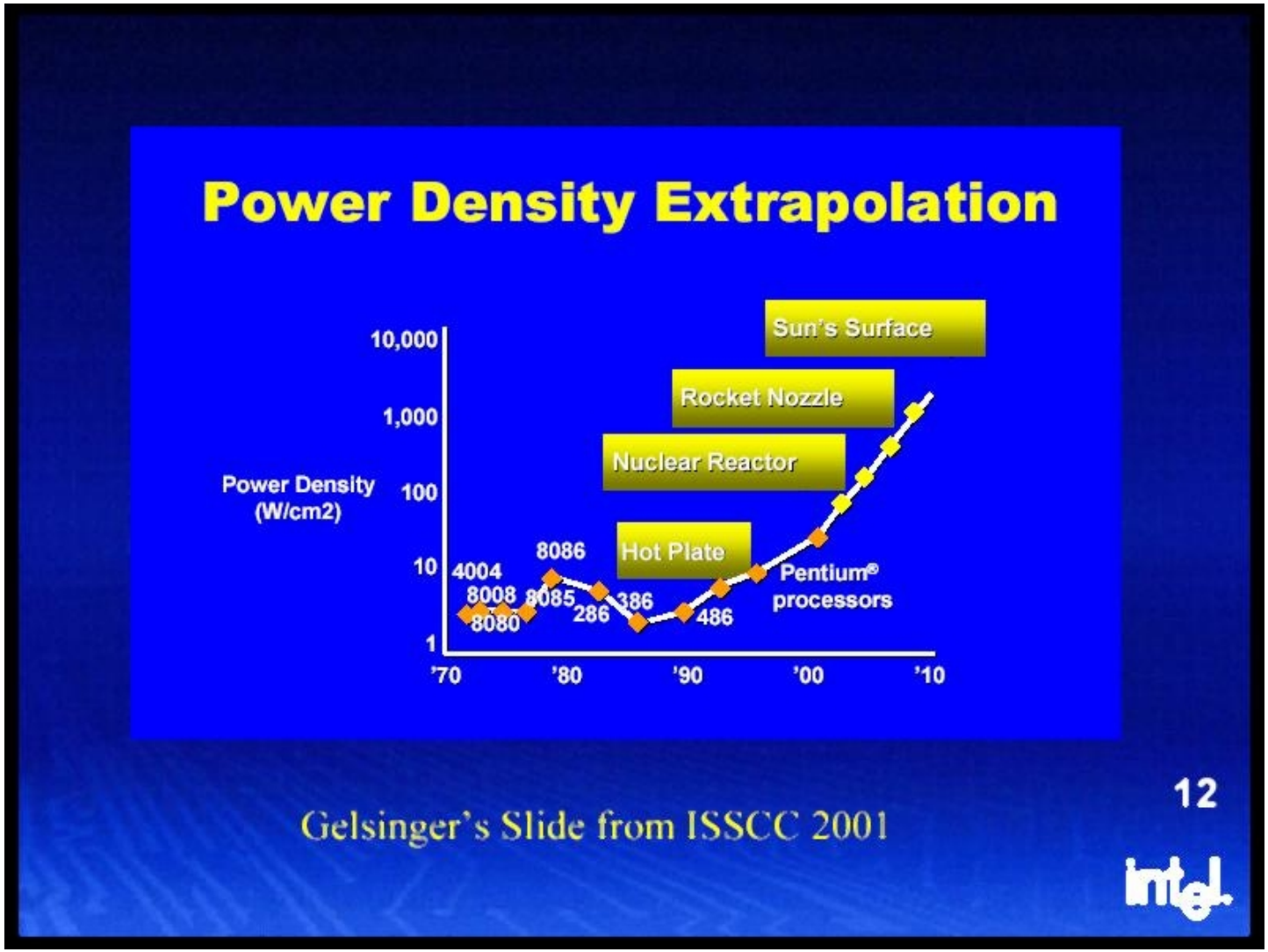
$$\text{Power} = \text{Capacitive load} \times \text{Voltage}^2 \times \text{Frequency}$$

×30

5V → 1V

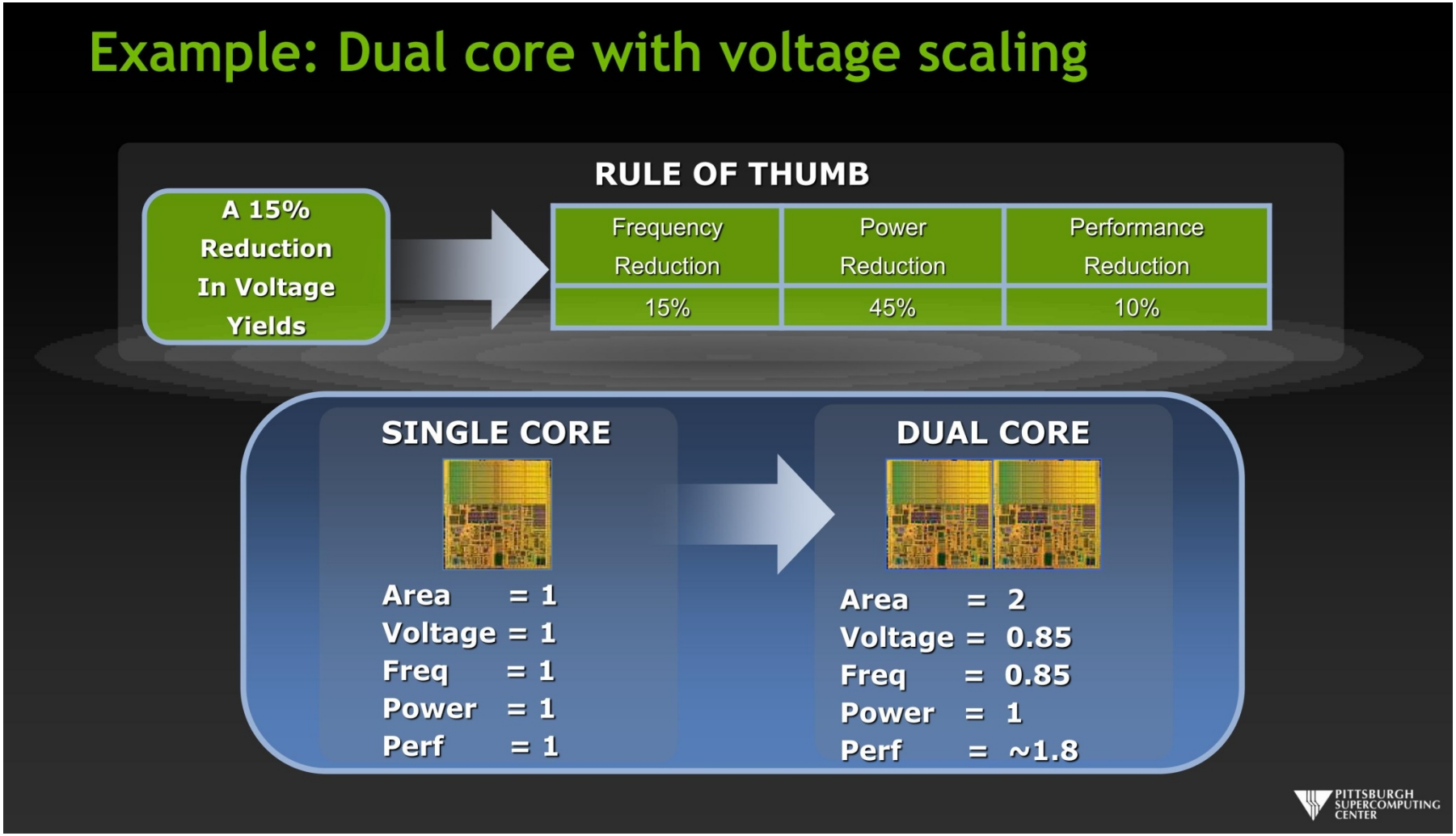
×1000

Power Density



Intel ISSCC 2001

Multiprocessor as the solution



B. Ganesan, Intel Corporation

Amdahl's Law

- Improving an aspect of a computer and expecting a proportional improvement in overall performance

$$T_{\text{improved}} = \frac{T_{\text{affected}}}{\text{improvement factor}} + T_{unaffected}$$

- Corollary: make the common case fast

Amdahl’s Law Example

A program can be split up into two parts:

A part which cannot be parallelized

A part which can be parallelized

Then if T = Total time of serial execution

B = Total time of non-parallizable part

T - B = Total time of parallizable part (when executed serially, not in parallel)

From this follows that:

T = B + (T-B)

The number of threads or CPUs is called N. The fastest the the parallelizable part can be executed is thus:(T - B) / N

Another way to write this is:

(1/N) * (T - B)

According to Amdahl's law, the total execution time of the program when the parallelizable part is executed using N threads or CPUs is thus:

$$T(N) = B + (T - B) / N$$

Amdahl's Law Example

The total time to execute a program is set to 1.

The non-parallelizable part of the programs is 40% which out of a total time of 1 is equal to 0.4

The parallelizable part is thus equal to 1 - 0.4 = 0.6 .

The execution time of the program with a parallelization factor of 2 (2 threads or CPUs executing the parallelizable part, so N is 2) would be:

T(2) = 0.4 + (1 - 0.4) / 2

= 0.4 + 0.6 / 2

= 0.4 + 0.3

= 0.7

Making the same calculation with a parallelization factor of 5 instead of 2 would look like this:

T(5) = 0.4 + (1 - 0.4) / 5

= 0.4 + 0.6 / 5

= 0.4 + 0.12

= 0.52